Week 4: Transformations and Expectations

Week 4: Transformations and Expectations

Reading: Casella & Berger (C&B) §§2.1–2.2

Lecture	Date	Topics	C&B
4.1	Tue. Oct. 20	Distributions of functions of a random variable: the discrete case, the cdf method, monotone transformations, piecewise-monotone transformations	§2.1
4.2	Thu. Oct. 22	The probability integral transformation; expected values; linearity and properties of expectation; EX as the best predictor	§§2.1–2.2

Notation.

- $Y = g(X)$ is a function of the random variable X.
- $\mathcal{X}$ and $\mathcal{Y}$ are the support sets of X and Y, as in (2.1.7).
- g^{-1} is the inverse mapping, defined on sets in (2.1.1).
- $\mathrm{E}\, g(X)$ is the expected value of $g(X)$.

Where we are. In Week 3 we learned to describe a single random variable X by its cdf F_X and its pmf or pdf f_X. This week answers two natural follow-up questions:

1. If we know the distribution of X, what is the distribution of $Y = g(X)$?
2. What single number summarizes the “center” of a distribution?

Distributions of Functions of a Random Variable

> **Way of thinking.** Don't transform the density; transform the *event*.

1. The inverse mapping and the discrete case

If X is a random variable, then any function $g(X)$ is also a random variable (it is still a function from S into the real numbers). Writing $y = g(x)$, the function $g(x)$ defines a mapping from the sample space $\mathcal{X}$ of X to a new sample space $\mathcal{Y}$, the sample space of Y:

$$g(x) : \mathcal{X} \to \mathcal{Y}.$$

Support sets. To avoid confusion, always keep track of where the random variables live. For $Y = g(X)$ it is most convenient to use

$$\begin{aligned} &\mathcal{X} = \{x : f_X(x) > 0\} \quad \text{and} \\ &\mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}. \end{aligned} \tag{2.1.7}$$

The set where a pdf (or pmf, or any nonnegative function) is positive is called its **support set**, or simply its **support**.

We associate with g an inverse mapping g^{-1}, which maps subsets of $\mathcal{Y}$ to subsets of $\mathcal{X}$:

$$g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}. \tag{2.1.1}$$

About g^{-1}:

- It takes **sets into sets**.
- It exists even when g is not one-to-one.
- For a point set, we write $g^{-1}(y) = \{x \in \mathcal{X} : g(x) = y\}$.
  - This set can contain several points.
  - If it contains exactly one point x, we write $g^{-1}(y) = x$.

The distribution of $Y = g(X)$ is then determined by that of X: for any set $A \subset \mathcal{Y}$,

$$\begin{aligned} P(Y \in A) &= P(g(X) \in A) \\ &= P\big(\{x \in \mathcal{X} : g(x) \in A\}\big) \\ &= P\big(X \in g^{-1}(A)\big). \end{aligned} \tag{2.1.2}$$

This is the same idea as the induced probability (1.4.2) from Week 3, applied one level up.

Discrete case. If X is discrete, then $\mathcal{X}$ is countable, so $\mathcal{Y} = \{y : y = g(x),\ x \in \mathcal{X}\}$ is countable and Y is discrete. From (2.1.2), the pmf of Y is

$$\begin{aligned} f_Y(y) = P(Y = y) &= \sum_{x \in g^{-1}(y)} P(X = x) \\ &= \sum_{x \in g^{-1}(y)} f_X(x), \quad \text{for } y \in \mathcal{Y}, \end{aligned}$$

and $f_Y(y) = 0$ for $y \notin \mathcal{Y}$. So, for each $y \in \mathcal{Y}$:

1. identify the set $g^{-1}(y)$;
2. add up $f_X(x)$ over the points x in that set.

Example 2.1.1 (Binomial transformation). A discrete random variable X has a **binomial distribution** if its pmf is of the form

$$f_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n, \tag{2.1.3}$$

where n is a positive integer and $0 \leq p \leq 1$. Values such as n and p that can be set to different values, producing different probability distributions, are called **parameters**.

Consider $Y = g(X)$ with $g(x) = n - x$, so $Y = n - X$.

- Here $\mathcal{X} = \mathcal{Y} = \{0, 1, \ldots, n\}$.
- $g^{-1}(y)$ is the single point $x = n - y$.

So

$$f_Y(y) = f_X(n-y) = \binom{n}{n-y} p^{n-y}(1-p)^y = \binom{n}{y}(1-p)^y p^{n-y},$$

using $\binom{n}{y} = \binom{n}{n-y}$ (Definition 1.2.17).

- Thus Y is also binomial, with parameters n and $1 - p$.
- Interpretation: if X counts heads, Y counts tails.

2. The cdf method

For continuous X, the cdf of $Y = g(X)$ is

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) \\ &= P\big(\{x \in \mathcal{X} : g(x) \leq y\}\big) \\ &= \int_{\{x \in \mathcal{X}:\, g(x) \leq y\}} f_X(x)\, dx. \end{aligned} \tag{2.1.4}$$

The hard part is identifying the set $\{x \in \mathcal{X} : g(x) \leq y\}$.

Example 2.1.2 (Uniform transformation). Suppose X has a uniform distribution on $(0, 2\pi)$, that is, $f_X(x) = 1/(2\pi)$ for $0 < x < 2\pi$ and 0 otherwise. Consider $Y = \sin^2(X)$.

For $0 < y < 1$, the line at height y crosses $\sin^2(x)$ at four points $x_1 < x_2 < \pi < x_3 < x_4$ (C&B Figure 2.1.1), and

$$P(Y \le y) = P(X \le x_1) + P(x_2 \le X \le x_3) + P(X \ge x_4). \tag{2.1.5}$$

By the symmetry of $\sin^2(x)$ and the uniformity of X,

$$P(Y \le y) = 2P(X \le x_1) + 2P(x_2 \le X \le \pi), \tag{2.1.6}$$

where x_1 and x_2 are the two solutions to $\sin^2(x) = y$, $0 < x < \pi$.

- Even this simple-looking transformation leads to a cdf that takes some work.
- Carrying it through with $x_1 = \arcsin\sqrt{y}$ and $x_2 = \pi - x_1$ gives $F_Y(y) = \frac{2}{\pi}\arcsin\sqrt{y}$ for $0 < y < 1$ (see Exercise 2.5).

Support sets. To avoid confusion, always keep track of where the random variables live. For $Y = g(X)$ it is most convenient to use

$$\begin{aligned} &\mathcal{X} = \{x : f_X(x) > 0\} \quad \text{and} \\ &\mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}. \end{aligned} \tag{2.1.7}$$

The set where a pdf (or pmf, or any nonnegative function) is positive is called its **support set**, or simply its **support**.

3. Monotone transformations

g is **monotone** if it is either

- **increasing:** $u > v \Rightarrow g(u) > g(v)$, or
- **decreasing:** $u < v \Rightarrow g(u) > g(v)$.

A monotone g is *one-to-one and onto* from $\mathcal{X}$ to $\mathcal{Y}$, so g^{-1} is single-valued: $g^{-1}(y) = x$ if and only if $y = g(x)$. Then

$$\{x \in \mathcal{X} : g(x) \le y\} = \{x \in \mathcal{X} : x \le g^{-1}(y)\} \quad \text{if } g \text{ is increasing,} \tag{2.1.8}$$

$$\{x \in \mathcal{X} : g(x) \le y\} = \{x \in \mathcal{X} : x \ge g^{-1}(y)\} \quad \text{if } g \text{ is decreasing.} \tag{2.1.9}$$

(A decreasing function reverses inequalities: draw a graph to see why.)

Theorem 2.1.3. Let X have cdf $F_X(x)$, let $Y = g(X)$, and let $\mathcal{X}$ and $\mathcal{Y}$ be defined as in (2.1.7).

- **a.** If g is an increasing function on $\mathcal{X}$, $F_Y(y) = F_X\big(g^{-1}(y)\big)$ for $y \in \mathcal{Y}$.

- **b.** If g is a decreasing function on $\mathcal{X}$ and X is a continuous random variable, $F_Y(y) = 1 - F_X\big(g^{-1}(y)\big)$ for $y \in \mathcal{Y}$.

Why.

- **Increasing g:** by (2.1.8), $F_Y(y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y))$.
- **Decreasing g:** by (2.1.9), $F_Y(y) = P(X \ge g^{-1}(y)) = 1 - F_X(g^{-1}(y))$. The last step uses continuity: $P(X = g^{-1}(y)) = 0$.

Example 2.1.4 (Uniform-exponential relationship–I). Suppose $X \sim f_X(x) = 1$ if $0 < x < 1$ and 0 otherwise, the **uniform(0, 1) distribution**, so $F_X(x) = x$ for $0 < x < 1$. Let $Y = g(X) = -\log X$. Since

$$\frac{d}{dx} g(x) = \frac{d}{dx}(-\log x) = -\frac{1}{x} < 0, \quad \text{for } 0 < x < 1,$$

g is decreasing, and $\mathcal{Y} = (0, \infty)$.

- For $y > 0$, $y = -\log x$ implies $x = e^{-y}$, so $g^{-1}(y) = e^{-y}$.
- By Theorem 2.1.3(b),

$$F_Y(y) = 1 - F_X\big(g^{-1}(y)\big) = 1 - F_X(e^{-y}) = 1 - e^{-y}.$$

- Of course $F_Y(y) = 0$ for $y \le 0$.

Note that we only needed g to be monotone on $(0, 1)$, the support of X.

Theorem 2.1.5. Let X have pdf $f_X(x)$ and let $Y = g(X)$, where g is a monotone function. Let $\mathcal{X}$ and $\mathcal{Y}$ be defined by (2.1.7). Suppose that $f_X(x)$ is continuous on $\mathcal{X}$ and that $g^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$. Then the pdf of Y is given by

$$f_Y(y) = \begin{cases} f_X\big(g^{-1}(y)\big) \left| \dfrac{d}{dy} g^{-1}(y) \right| & y \in \mathcal{Y} \\ 0 & \text{otherwise.} \end{cases} \tag{2.1.10}$$

Proof. From Theorem 2.1.3, by the chain rule,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} f_X\big(g^{-1}(y)\big) \dfrac{d}{dy} g^{-1}(y) & \text{if } g \text{ is increasing} \\ -f_X\big(g^{-1}(y)\big) \dfrac{d}{dy} g^{-1}(y) & \text{if } g \text{ is decreasing,} \end{cases}$$

which can be expressed concisely as (2.1.10). □

Why the absolute value.

- $f_X(g^{-1}(y))$ says *where* y comes from.
- The factor $|dx/dy|$ says how much g stretches or squeezes a small interval.
- Probability is conserved: $f_Y(y)\,|dy| = f_X(x)\,|dx|$.
- Densities are never negative, so the factor must be $|dx/dy|$, not dx/dy.

Example 2.1.6 (Inverted gamma pdf). Let $f_X(x)$ be the **gamma pdf**

$$f(x) = \frac{1}{(n-1)!\,\beta^n} x^{n-1} e^{-x/\beta}, \quad 0 < x < \infty,$$

where β is a positive constant and n is a positive integer. To find the pdf of $g(X) = 1/X$, note that

- $\mathcal{X} = \mathcal{Y} = (0, \infty)$;
- $g^{-1}(y) = 1/y$;
- $\frac{d}{dy} g^{-1}(y) = -1/y^2$.

By Theorem 2.1.5, for $y \in (0, \infty)$,

$$\begin{aligned} f_Y(y) &= f_X\big(g^{-1}(y)\big) \left| \frac{d}{dy} g^{-1}(y) \right| = \frac{1}{(n-1)!\,\beta^n} \left(\frac{1}{y}\right)^{n-1} e^{-1/(\beta y)} \frac{1}{y^2} \\ &= \frac{1}{(n-1)!\,\beta^n} \left(\frac{1}{y}\right)^{n+1} e^{-1/(\beta y)}, \end{aligned}$$

a special case of the **inverted gamma pdf**.

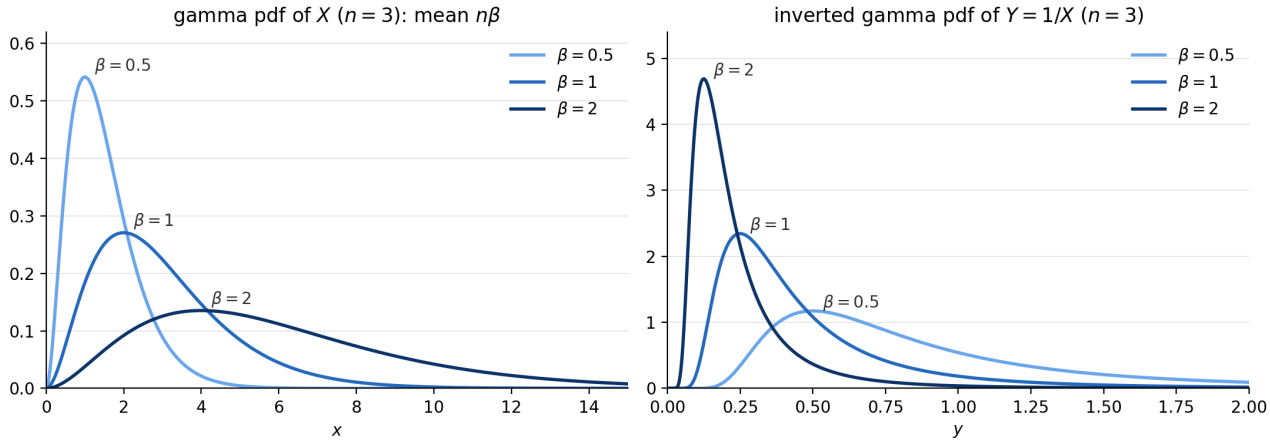


Figure. The gamma pdf of X (left) and the inverted gamma pdf of $Y = 1/X$ (right), with $n = 3$ and $\beta = 0.5, 1, 2$.

- **Left:** a larger β spreads X to the right. The mean is $n\beta$ and the mode is $(n-1)\beta$.
- **Right:** taking reciprocals reverses the order. A larger β pushes Y toward 0, with mode $1/\big((n+1)\beta\big)$.

- Large values of X become small values of Y, and the factor $1/y^2$ from the Jacobian reshapes the curve. f_Y is **not** simply $f_X(1/y)$.

4. Piecewise-monotone transformations

Often g is neither increasing nor decreasing, but it is monotone on pieces. (If g is not monotone over certain intervals, then, as C&B put it, "we are in *deep* trouble.")

Example 2.1.7 (Square transformation). Suppose X is a continuous random variable. For $y > 0$, the cdf of $Y = X^2$ is

$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y}).$$

Because X is continuous, we can drop the equality from the left endpoint:

$$\begin{aligned} F_Y(y) &= P(-\sqrt{y} < X \le \sqrt{y}) \\ &= P(X \le \sqrt{y}) - P(X \le -\sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}). \end{aligned}$$

Differentiating, with the chain rule for each term,

$$\begin{aligned} f_Y(y) &= \frac{d}{dy}\left[F_X(\sqrt{y}) - F_X(-\sqrt{y})\right] \\ &= \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}), \end{aligned}$$

so

$$f_Y(y) = \frac{1}{2\sqrt{y}}\left(f_X(\sqrt{y}) + f_X(-\sqrt{y})\right). \tag{2.1.11}$$

The pdf of Y is a sum of two pieces, one for each interval on which $g(x) = x^2$ is monotone. In general, this will be the case.

Theorem 2.1.8. Let X have pdf $f_X(x)$, let $Y = g(X)$, and define the sample space $\mathcal{X}$ as in (2.1.7). Suppose there exists a partition, $A_0, A_1, \ldots, A_k$, of $\mathcal{X}$ such that $P(X \in A_0) = 0$ and $f_X(x)$ is continuous on each A_i. Further, suppose there exist functions $g_1(x), \ldots, g_k(x)$, defined on $A_1, \ldots, A_k$, respectively, satisfying

- **i.** $g(x) = g_i(x)$, for $x \in A_i$,
- **ii.** $g_i(x)$ is monotone on A_i,
- **iii.** the set $\mathcal{Y} = \{y : y = g_i(x) \text{ for some } x \in A_i\}$ is the same for each $i = 1, \ldots, k$, and
- **iv.** $g_i^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$, for each $i = 1, \ldots, k$.

Then

$$f_Y(y) = \begin{cases} \displaystyle\sum_{i=1}^{k} f_X\big(g_i^{-1}(y)\big) \left| \frac{d}{dy} g_i^{-1}(y) \right| & y \in \mathcal{Y} \\ 0 & \text{otherwise.} \end{cases}$$

Reading the theorem.

- Split $\mathcal{X}$ into pieces $A_1, \ldots, A_k$ on which g is monotone.
- Each g_i is a one-to-one map from A_i onto $\mathcal{Y}$, so each y has exactly one "pre-image" $g_i^{-1}(y)$ in each piece.
- $f_Y(y)$ adds up the contributions from all the pieces.
- The "exceptional set" A_0 has probability 0. It is a technical device for handling endpoints such as $x = 0$ below.
- See Exercise 2.7 for an extension.

Example 2.1.9 (Normal-chi squared relationship). Let X have the **standard normal distribution**,

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty,$$

and let $Y = X^2$. The function $g(x) = x^2$ is monotone on $(-\infty, 0)$ and on $(0, \infty)$, and $\mathcal{Y} = (0, \infty)$. Take

$$\begin{aligned} &A_0 = \{0\}; \\ &A_1 = (-\infty, 0), && g_1^{-1}(y) = -\sqrt{y}; \\ &A_2 = (0, \infty), && g_2^{-1}(y) = \sqrt{y}. \end{aligned}$$

By Theorem 2.1.8,

$$\begin{aligned} f_Y(y) &= \frac{1}{\sqrt{2\pi}} e^{-(-\sqrt{y})^2/2} \left| -\frac{1}{2\sqrt{y}} \right| + \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{y})^2/2} \left| \frac{1}{2\sqrt{y}} \right| \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-y/2}, \quad 0 < y < \infty. \end{aligned}$$

This is the pdf of a **chi squared random variable with 1 degree of freedom**, which we will meet often.

Check your understanding.

1. Let $X \sim$ uniform(0, 1) and $Y = X^2$. Find F_Y and f_Y in two ways: by the cdf method, and by Theorem 2.1.5. Is f_Y bounded?

> On $(0, 1)$, $g(x) = x^2$ is increasing, so $F_Y(y) = P(X \le \sqrt{y}) = \sqrt{y}$ for $0 < y < 1$. By Theorem 2.1.5 with $g^{-1}(y) = \sqrt{y}$: $f_Y(y) = 1 \cdot \frac{1}{2\sqrt{y}}$, $0 < y < 1$, which matches $\frac{d}{dy}\sqrt{y}$. It is **unbounded** near $y = 0$, yet still integrates to 1: a pdf is not a probability.

2. Let X have pdf f_X and $Y = aX + b$ with $a \neq 0$. Show that $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$.

> $g(x) = ax + b$ is monotone (increasing if $a > 0$, decreasing if $a < 0$), with $g^{-1}(y) = (y-b)/a$ and $\frac{d}{dy} g^{-1}(y) = 1/a$. Theorem 2.1.5 gives the formula. This is the basis of the location–scale families in Chapter 3.

3. Let X be the number of heads in 3 tosses of a fair coin and $Y = (X-1)^2$. Find the pmf of Y.

> $g^{-1}(0) = \{1\}$, $g^{-1}(1) = \{0, 2\}$, $g^{-1}(4) = \{3\}$. So $f_Y(0) = 3/8$, $f_Y(1) = 1/8 + 3/8 = 1/2$, and $f_Y(4) = 1/8$. Two values of x map to $y = 1$, so their probabilities add.

The Probability Integral Transformation and Expected Values

> **Way of thinking.** The expected value is a probability-weighted average.

5. The probability integral transformation

Theorem 2.1.10 (Probability integral transformation). Let X have continuous cdf $F_X(x)$ and define the random variable Y as $Y = F_X(X)$. Then Y is uniformly distributed on $(0, 1)$, that is, $P(Y \leq y) = y$, $0 < y < 1$.

The inverse cdf.

- If F_X is strictly increasing, then F_X^{-1} is well defined by

$$F_X^{-1}(y) = x \iff F_X(x) = y. \tag{2.1.12}$$

- But if F_X is constant on some interval $[x_1, x_2]$, then every x in that interval satisfies $F_X(x) = y$ (C&B Figure 2.1.2b), and (2.1.12) does not define a single value.

- This is avoided by defining, for $0 < y < 1$,

$$F_X^{-1}(y) = \inf\{x : F_X(x) \geq y\}. \tag{2.1.13}$$

  This agrees with (2.1.12) when F_X is nonconstant, and it is single-valued even when F_X is not strictly increasing.

- At the endpoints: $F_X^{-1}(1) = \infty$ if $F_X(x) < 1$ for all x, and, for any F_X, $F_X^{-1}(0) = -\infty$.

Proof of Theorem 2.1.10. For $Y = F_X(X)$ and $0 < y < 1$,

$$\begin{aligned} P(Y \le y) &= P\big(F_X(X) \le y\big) \\ &= P\big(F_X^{-1}[F_X(X)] \le F_X^{-1}(y)\big) && (F_X^{-1} \text{ is increasing}) \\ &= P\big(X \le F_X^{-1}(y)\big) && (\text{see below}) \\ &= F_X\big(F_X^{-1}(y)\big) && (\text{definition of } F_X) \\ &= y. && (\text{continuity of } F_X) \end{aligned}$$

At the endpoints, $P(Y \le y) = 1$ for $y \ge 1$ and $P(Y \le y) = 0$ for $y \le 0$, so Y is uniform(0, 1).

The third equality deserves attention:

- If F_X is strictly increasing, $F_X^{-1}(F_X(x)) = x$.
- If F_X is flat on $[x_1, x_2]$, then $F_X^{-1}(F_X(x)) = x_1$ for any x in that interval, which is not x.
- But the probability equality still holds, because $P(X \le x) = P(X \le x_1)$ for any $x \in [x_1, x_2]$. The flat part of the cdf has probability 0: $P(x_1 < X \le x) = F_X(x) - F_X(x_1) = 0$. □

Application: simulation. To generate an observation X with cdf F_X:

1. generate a uniform(0, 1) random number V;
2. solve $F_X(x) = V$ for x, that is, set $X = F_X^{-1}(V)$.

For example, by Example 2.1.4, $-\log V$ has the exponential distribution with $F(y) = 1 - e^{-y}$.

- This "inverse cdf method" is how computers sample from many distributions.
- It also gives a concrete answer to Theorem 1.5.3's claim that every cdf belongs to *some* random variable.

6. Expected values

The expected value, or expectation, of a random variable is its average value, where "average" is weighted according to the probability distribution. It is a measure of center.

Definition 2.2.1. The **expected value** or **mean** of a random variable $g(X)$, denoted by $\mathrm{E}\, g(X)$, is

$$\mathrm{E}\, g(X) = \begin{cases} \displaystyle\int_{-\infty}^{\infty} g(x) f_X(x)\, dx & \text{if } X \text{ is continuous} \\ \displaystyle\sum_{x \in \mathcal{X}} g(x) f_X(x) = \sum_{x \in \mathcal{X}} g(x) P(X = x) & \text{if } X \text{ is discrete,} \end{cases}$$

provided that the integral or sum exists. If $\mathrm{E}|g(X)| = \infty$, we say that $\mathrm{E}\, g(X)$ does not exist.

Notice that we compute $\mathrm{E}\, g(X)$ using the distribution of X, without first finding the distribution of $g(X)$. (Ross 1988 calls this the "law of the unconscious statistician"; C&B "do not find this amusing.")

Example 2.2.2 (Exponential mean). Suppose X has an **exponential(λ) distribution**, with pdf

$$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}, \quad 0 \le x < \infty, \quad \lambda > 0.$$

Then, integrating by parts,

$$\begin{aligned} \mathrm{E}X &= \int_0^\infty \frac{1}{\lambda} x e^{-x/\lambda}\, dx \\ &= -x e^{-x/\lambda}\Big|_0^\infty + \int_0^\infty e^{-x/\lambda}\, dx \\ &= \int_0^\infty e^{-x/\lambda}\, dx = \lambda. \end{aligned}$$

Example 2.2.3 (Binomial mean). If $X \sim$ binomial(n, p) as in (2.1.3), then (the $x = 0$ term is 0)

$$\mathrm{E}X = \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^n x \binom{n}{x} p^x (1-p)^{n-x}.$$

Using the identity $x\binom{n}{x} = n\binom{n-1}{x-1}$ and substituting $y = x - 1$,

$$\begin{aligned} \mathrm{E}X &= \sum_{x=1}^n n \binom{n-1}{x-1} p^x (1-p)^{n-x} \\ &= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{n-1-y} = np, \end{aligned}$$

since the last sum is the total of a binomial($n-1$, p) pmf, which is 1. (Trick worth remembering: factor out constants until what remains is a pmf or pdf that sums or integrates to 1.)

Example 2.2.4 (Cauchy mean). A classic random variable whose expected value does not exist is a **Cauchy random variable**, with pdf

$$f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

It is straightforward to check that $\int_{-\infty}^\infty f_X(x)\, dx = 1$, but $\mathrm{E}|X| = \infty$. For any positive number M,

$$\int_0^M \frac{x}{1+x^2}\, dx = \frac{\log(1+x^2)}{2}\Big|_0^M = \frac{\log(1+M^2)}{2},$$

so

$$\begin{aligned}\mathrm{E}|X| &= \int_{-\infty}^{\infty} \frac{|x|}{\pi}\,\frac{1}{1+x^2}\,dx = \frac{2}{\pi}\int_{0}^{\infty} \frac{x}{1+x^2}\,dx \\ &= \lim_{M\to\infty} \frac{1}{\pi}\log(1+M^2) = \infty,\end{aligned}$$

and $\mathrm{E}X$ does not exist.

- The density is symmetric about 0, but "the positive and negative parts cancel" is not allowed.
- Both parts are infinite, and $\infty - \infty$ is undefined.

7. Properties of expectation

Taking expectations is a **linear** operation: for any constants a and b,

$$\mathrm{E}(aX + b) = a\,\mathrm{E}X + b. \tag{2.2.1}$$

For example, if $X \sim$ binomial(n, p), then $\mathrm{E}(X - np) = \mathrm{E}X - np = 0$.

Theorem 2.2.5. Let X be a random variable and let a, b, and c be constants. Then for any functions $g_1(x)$ and $g_2(x)$ whose expectations exist,

- **a.** $\mathrm{E}\big(ag_1(X) + bg_2(X) + c\big) = a\,\mathrm{E}\,g_1(X) + b\,\mathrm{E}\,g_2(X) + c$.
- **b.** If $g_1(x) \geq 0$ for all x, then $\mathrm{E}\,g_1(X) \geq 0$.
- **c.** If $g_1(x) \geq g_2(x)$ for all x, then $\mathrm{E}\,g_1(X) \geq \mathrm{E}\,g_2(X)$.
- **d.** If $a \leq g_1(x) \leq b$ for all x, then $a \leq \mathrm{E}\,g_1(X) \leq b$.

Proof of (a), continuous case. By the additivity of the integral, and since constants factor out,

$$\begin{aligned}\mathrm{E}\big(ag_1(X) + bg_2(X) + c\big) &= \int_{-\infty}^{\infty} \big(ag_1(x) + bg_2(x) + c\big) f_X(x)\,dx \\ &= a\int_{-\infty}^{\infty} g_1(x) f_X(x)\,dx + b\int_{-\infty}^{\infty} g_2(x) f_X(x)\,dx \\ &\quad + c\int_{-\infty}^{\infty} f_X(x)\,dx,\end{aligned}$$

which is $a\,\mathrm{E}\,g_1(X) + b\,\mathrm{E}\,g_2(X) + c$. The discrete case and parts (b)–(d) are similar. □

Caution. Linearity does **not** extend to nonlinear functions: in general $\mathrm{E}\,g(X) \neq g(\mathrm{E}X)$. For example, if $X \sim$ uniform(0, 1), then $\mathrm{E}X^2 = 1/3 \neq (\mathrm{E}X)^2 = 1/4$.

Example 2.2.6 (Minimizing distance). Measure the distance between X and a constant b by $(X - b)^2$. Which b minimizes $\mathrm{E}(X - b)^2$? (Minimizing $(X - b)^2$ itself is useless, since the answer would depend on X.) Adding and subtracting $\mathrm{E}X$,

$$\begin{aligned}\mathrm{E}(X-b)^2 &= \mathrm{E}(X-\mathrm{E}X+\mathrm{E}X-b)^2 \\ &= \mathrm{E}\big((X-\mathrm{E}X)+(\mathrm{E}X-b)\big)^2 \\ &= \mathrm{E}(X-\mathrm{E}X)^2+(\mathrm{E}X-b)^2+2\,\mathrm{E}\big((X-\mathrm{E}X)(\mathrm{E}X-b)\big).\end{aligned}$$

The last term is $(\mathrm{E}X-b)\,\mathrm{E}(X-\mathrm{E}X)=(\mathrm{E}X-b)(\mathrm{E}X-\mathrm{E}X)=0$, because $\mathrm{E}X-b$ is a constant. So

$$\mathrm{E}(X-b)^2=\mathrm{E}(X-\mathrm{E}X)^2+(\mathrm{E}X-b)^2. \tag{2.2.2}$$

- We cannot control the first term.
- The second term is ≥ 0, with equality when $b=\mathrm{E}X$.

Hence

$$\min_b \mathrm{E}(X-b)^2=\mathrm{E}(X-\mathrm{E}X)^2. \tag{2.2.3}$$

- $\mathrm{E}X$ is the **best constant predictor** of X under squared error.
- Exercise 2.19 gives a calculus proof; Exercise 2.18 gives the analogous result for the median under absolute error.
- The minimum value, $\mathrm{E}(X-\mathrm{E}X)^2$, is the variance, which starts next week.

Two ways to compute $\mathrm{E}\,g(X)$.

1. Directly from the definition:

$$\mathrm{E}\,g(X)=\int_{-\infty}^{\infty} g(x)f_X(x)\,dx. \tag{2.2.4}$$

2. By first finding the pdf f_Y of $Y=g(X)$ (Section 2.1), and then

$$\mathrm{E}\,g(X)=\mathrm{E}Y=\int_{-\infty}^{\infty} y f_Y(y)\,dy. \tag{2.2.5}$$

Example 2.2.7 (Uniform-exponential relationship–II). Let $X\sim$ uniform(0, 1), with $f_X(x)=1$ if $0\leq x\leq 1$ and 0 otherwise, and let $g(X)=-\log X$. By (2.2.4),

$$\mathrm{E}\,g(X)=\mathrm{E}(-\log X)=\int_0^1 -\log x\,dx=\big(x-x\log x\big)\Big|_0^1=1.$$

Alternatively, by (2.2.5):

- By Example 2.1.4, $Y=-\log X$ has cdf $1-e^{-y}$.
- Hence its pdf is $f_Y(y)=\frac{d}{dy}(1-e^{-y})=e^{-y}$, $0<y<\infty$, the exponential pdf with $\lambda=1$.
- So by Example 2.2.2, $\mathrm{E}Y=1$, the same answer.

Check your understanding.

1. Let X have pmf $P(X=-1)=0.2$, $P(X=0)=0.5$, $P(X=2)=0.3$. Find $\mathrm{E}X$, $\mathrm{E}X^2$, and $\mathrm{E}(3X-1)$. Is $\mathrm{E}X^2=(\mathrm{E}X)^2$?

   > $\mathrm{E}X=-0.2+0+0.6=0.4$. $\mathrm{E}X^2=0.2+0+1.2=1.4$. $\mathrm{E}(3X-1)=3(0.4)-1=0.2$ by (2.2.1). $\mathrm{E}X^2=1.4\neq 0.16=(\mathrm{E}X)^2$.

2. Let X be continuous with cdf F_X. What is the distribution of $1-F_X(X)$? Use this to explain why $\mathrm{E}F_X(X)=1/2$.

   > By Theorem 2.1.10, $U=F_X(X)\sim$ uniform(0, 1). Then $1-U$ is also uniform(0, 1) (Check your understanding 2 of Lecture 4.1 with $a=-1$, $b=1$). And $\mathrm{E}F_X(X)=\mathrm{E}U=\int_0^1 u\,du=1/2$.

3. Does the expected value of the pdf $f(x)=1/x^2$, $1<x<\infty$, exist?

   > f is a pdf: $\int_1^\infty x^{-2}\,dx=1$. But $\mathrm{E}X=\int_1^\infty x\cdot x^{-2}\,dx=\int_1^\infty \frac{dx}{x}=\infty$, so the mean does not exist (here it is $+\infty$, since $X>0$). Heavy tails can make the mean infinite, as in the Cauchy example.

Summary of Week 4

- **Functions of a random variable:** $P(Y\in A)=P(X\in g^{-1}(A))$ (2.1.2). Discrete case: add $f_X(x)$ over $x\in g^{-1}(y)$.
- **Cdf method:** $F_Y(y)=P(g(X)\leq y)$; identify the x-set, then differentiate. Keep track of the supports $\mathcal{X}$ and $\mathcal{Y}$ (2.1.7).
- **Monotone g:** $F_Y(y)=F_X(g^{-1}(y))$ or $1-F_X(g^{-1}(y))$ (Theorem 2.1.3), and $f_Y(y)=f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$ (Theorem 2.1.5). Piecewise monotone: sum over the pieces (Theorem 2.1.8).
- **Probability integral transformation:** $F_X(X)\sim$ uniform(0, 1) for continuous F_X; conversely, $F_X^{-1}(V)$ with $V\sim$ uniform(0, 1) has cdf F_X.
- **Expectation:** $\mathrm{E}\,g(X)=\int g(x)f_X(x)\,dx$ or $\sum g(x)f_X(x)$, if $\mathrm{E}|g(X)|<\infty$; it may not exist (Cauchy). It is linear (Theorem 2.2.5), but $\mathrm{E}\,g(X)\neq g(\mathrm{E}X)$ in general. $\mathrm{E}X$ minimizes $\mathrm{E}(X-b)^2$.

Practice Problems

From C&B §2.5:

- **Transformations (discrete and monotone):** 2.1, 2.2, 2.3.
- **Piecewise-monotone transformations:** 2.5, 2.6, 2.7.
- **Inverse cdfs and the probability integral transformation:** 2.8, 2.9, 2.10.
- **Expected values:** 2.11, 2.12, 2.13, 2.14, 2.15, 2.17, 2.18, 2.19, 2.20.

Next week: moments and moment generating functions; differentiating under an integral sign (§§2.3–2.4).